

ORF 245 Midterm Exam Solutions

Fall 2023

Part I

1. False. It's possible to observe a random sample with very small (even 0) variance.
2. True. $P(A \cup B) \geq P(A) \geq P(A)P(B)$ since $P(B) \leq 1$.
3. False. Let $B = A$ for some A with $0 < P(A) < 1$. Then $P(A \cap B) = P(A) > P(A)P(B)$.
4. $10^4 = 10,000$
5. True. $P(n, k) = \frac{n!}{(n-k)!} > \frac{n!}{k!(n-k)!} = C(n, k)$
6. $(100)(99)(98)$
7. $P(A|B) = 0$ since $P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(\emptyset)}{(1/2)} = 0$
8. False. $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{2}$ but $P(A \cap B) = \frac{1}{6}$.
9. n since the number of successes in trials is at most n .
10. range = $\{0, 1, 2, \dots\}$ by the definition/pmf.

Part I:

11- False. For one, a Gaussian r.v. has range $\mathbb{R}$, not $\mathbb{Z}^+$. In addition, number of "arrivals" within an interval of time are well-modeled by Poisson r.v.'s.

12- False. Consider $X \sim \text{Unit}[0, \frac{1}{2}]$, whose density is $f(x) = \begin{cases} 2 & \text{if } x \in [0, \frac{1}{2}] \\ 0 & \text{o/w} \end{cases}$.

13- True. The joint density is non-negative (since f_x and f_y are), and

$$\begin{aligned} \iint f(x, y) dx dy &= \iint f_x(x) f_y(y) dx dy \\ &= \int f_y(y) \cdot \underbrace{\left(\int f_x(x) dx \right)}_{=1} dy \\ &= \int f_y(y) \cdot 1 dy = 1 \end{aligned}$$

14- True. Expectation is a linear operator (because integrals are)

15- True. Independence implies that $f(x, y) = f_x(x) f_y(y)$, so

$$\begin{aligned} E[XY] &= \iint xy f(x, y) dx dy \\ &= \int y f_y(y) \underbrace{\left(\int x f_x(x) dx \right)}_{E[X]} dy \\ &= E[X] \cdot \underbrace{\int y f_y(y) dy}_{= E[Y]} \end{aligned}$$

16- Let $Y = X^2$. $\text{Var}(Y) = E[Y^2] - E[Y]^2$ So:

$$\text{Var}(X^2) = E[X^4] - E[X^2]^2$$

$$\leadsto \underline{\underline{E[X^4] = \text{Var}(X^2) + E[X^2]^2}}$$

17- $f(x) = \frac{1}{\sqrt{2\pi \cdot 4}} e^{-\frac{(x+1)^2}{2 \cdot 4}}$

18- $IP(-5 \leq X \leq 1) = IP(-4 \leq X+1 \leq 2)$
 $= IP(-2 \leq \frac{X+1}{2} \leq 1) \quad \left. \vphantom{IP(-2 \leq \frac{X+1}{2} \leq 1)} \right\} \frac{X+1}{2} \sim N(0,1)$
 $= \underline{\underline{\Phi(1) - \Phi(-2)}}$

19- True. $\rho_{XY} = \frac{\text{Cov}(X,Y)}{\sqrt{\text{Var}(X) \cdot \text{Var}(Y)}}$, equals 0 if $\text{Cov}(X,Y) = 0$.

* Technically we need $\text{Var}(X) > 0$ and $\text{Var}(Y) > 0$ too... *

20- True. $\text{Cov}(-X, Y) = -\text{Cov}(X, Y)$, while $\text{Var}(X) = \text{Var}(-X)$.

Part II

1. (a) Let $x \in (A \cap B)^c$. Then $x \notin A \cap B$. Therefore, either $x \notin A$ (so $x \in A^c$) or $x \notin B$ (so $x \in B^c$) or both. In either case, $x \in A^c \cup B^c$. Hence, $(A \cap B)^c \subset A^c \cup B^c$.

(b) Let $x \in A^c \cup B^c$. Then $x \in A^c$ or $x \in B^c$ (or both). If $x \in A^c$ then $x \notin A$ so $x \notin A \cap B$ and so $x \in (A \cap B)^c$. If $x \in B^c$ then $x \notin B$ so $x \notin A \cap B$, and again $x \in (A \cap B)^c$. Hence, $A^c \cup B^c \subset (A \cap B)^c$.

(c) False. Many examples will work to show this. For example, if $A = S$ and $B = \emptyset$. Then $A \cup B^c = S$ but $B \cup A^c = \emptyset$. For a more concrete example, let $S = \{0, 1\}$, $A = \{0\}$, and $B = \{1\}$. Then $A \cup B^c = \{0\}$ but $B \cup A^c = \{1\}$.

(d) For any B , $B \cup B^c = S$. Therefore,
$$A \cap (B \cup B^c) = A \cap S = A.$$

2. (a) Let R_1 = event that first marble is red.

B_0 = event that Bag 0 is selected.

B_1 = event that Bag 1 is selected.

From the Law of Total Probability,

$$\begin{aligned} P(R_1) &= P(R_1|B_0)P(B_0) + P(R_1|B_1)P(B_1) \\ &= (1)\left(\frac{1}{2}\right) + \left(\frac{2}{3}\right)\left(\frac{1}{2}\right) = \frac{5}{6} \end{aligned}$$

(b) Let R_2 = event that second marble is red

$$\begin{aligned} P(R_1 \cap R_2) &= P(R_1 \cap R_2|B_0)P(B_0) + P(R_1 \cap R_2|B_1)P(B_1) \\ &= (1)\left(\frac{1}{2}\right) + \left(\frac{1}{3}\right)\left(\frac{1}{2}\right) = \frac{4}{6} = \frac{2}{3} \end{aligned}$$

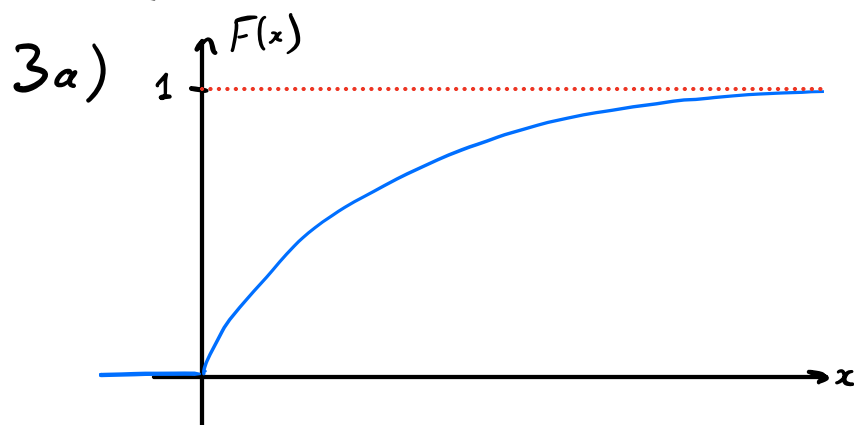
Note: Can see $P(R_1 \cap R_2|B_1) = \frac{1}{3}$ either by noting with prob. $\frac{1}{3}$ the white marble is the one not chosen,

$$\begin{aligned} \text{or } P(R_1 \cap R_2|B_1) &= P(R_1|B_1)P(R_2|R_1, B_1) \\ &= \left(\frac{2}{3}\right)\left(\frac{1}{2}\right) = \frac{1}{3} \end{aligned}$$

(c) Using Bayes' Rule,

$$\begin{aligned} P(B_0|R_1 \cap R_2) &= \frac{P(R_1 \cap R_2|B_0)P(B_0)}{P(R_1 \cap R_2)} \\ &= \frac{(1)\left(\frac{1}{2}\right)}{\left(\frac{2}{3}\right)} = \frac{3}{4} \end{aligned}$$

Part II:



$$\begin{aligned} 3b) \quad P(X \geq t) &= 1 - P(X < t) \\ &= 1 - P(X \leq t) \end{aligned} \quad \left. \begin{array}{l} \text{Since } X \text{ is cts., } P(X=t)=0 \end{array} \right\}$$

$$\begin{aligned} &= 1 - F(t) \\ \underline{P(X \geq t)} &= e^{-\lambda t} \end{aligned} \quad \left. \begin{array}{l} F(t) = \int_0^t \lambda \cdot e^{-\lambda u} du = \lambda \cdot \frac{-1}{\lambda} [e^{-\lambda u}]_0^t = 1 - e^{-\lambda t} \end{array} \right\}$$

3c) By our def'n of conditional density,

$$\begin{aligned} f_{X|X \geq t}(x) &= \frac{f(x)}{P(X \geq t)} \\ &= \frac{\lambda e^{-\lambda x}}{e^{-\lambda t}} \end{aligned}$$

$$f_{X|X \geq t}(x) = \lambda \cdot e^{-\lambda(x-t)} \quad \underline{\text{only}} \text{ for } x \geq t!$$

$$\text{So: } \underline{f_{X|X \geq t}(x) = \begin{cases} \lambda \cdot e^{-\lambda(x-t)} & \text{if } x \geq t \\ 0 & \text{otherwise} \end{cases}}$$

$$3d) \quad E[X|X \geq t] = \int_0^{\infty} x \cdot f_{X|X \geq t}(x) dx$$

$$\begin{aligned} &= \int_t^{\infty} x \cdot \lambda \cdot e^{-\lambda(x-t)} dx \\ &= \int_0^{\infty} (u+t) \cdot \lambda \cdot e^{-\lambda u} du \end{aligned} \quad \left. \begin{array}{l} \text{Substitution: } u = x-t \\ du = dx \end{array} \right\}$$

$$= \underbrace{\int_0^{\infty} u \cdot \lambda \cdot e^{-\lambda u} du}_{=E[X]=\frac{1}{\lambda}} + t \cdot \underbrace{\int_0^{\infty} \lambda \cdot e^{-\lambda u} du}_{=1, \text{ since } \lambda \cdot e^{-\lambda u} \text{ is a density}}$$

$$\underline{E[X|X \geq t] = t + \frac{1}{\lambda}}$$

3e) Think of X as being a "waiting time". Part d) tells us that at any point in time t , my expected additional wait is the same (namely equal to $1/\lambda$).

So, the idea of "memoryless" is that the distribution of $X|X \geq t$ is the same as that of $t+X$.

$$4a) \quad P_X(x) = \begin{cases} 0.5 = 0.1 + 0.1 + 0.3 & \text{if } x=0 \\ 0.2 = 0.1 + 0.1 & \text{if } x=1 \\ 0.3 = 0.2 + 0.1 & \text{if } x=2 \\ 0 & \text{otherwise} \end{cases}, \quad P_Y(y) = \begin{cases} 0.4 & \text{if } y=0 \\ 0.2 & \text{if } y=1 \\ 0.4 & \text{if } y=2 \\ 0 & \text{otherwise} \end{cases}$$

4b) No. This follows from $x=1$ and $y=2$.

$$f(1,2) = 0, \text{ but } f_X(1) \cdot f_Y(2) = 0.2 \cdot 0.4 \neq 0$$

$$4c) \quad P_{X|Y=2}(x) = \begin{cases} 3/4 = \frac{0.3}{0.3+0.1} & \text{if } x=0 \\ 0 & \text{if } x=1 \\ 1/4 = \frac{0.1}{0.3+0.1} & \text{if } x=2 \\ 0 & \text{otherwise} \end{cases}$$

$$4d) \quad E[X] = 0 \cdot (0.5) + 1 \cdot (0.2) + 2 \cdot (0.3) = 0.8$$

$$E[Y] = 0 \cdot (0.4) + 1 \cdot (0.2) + 2 \cdot (0.4) = 1$$

Now:

$$E[XY] = 0.1(1 \cdot 1) + 0.1(2 \cdot 2) + 0$$

$$E[XY] = 0.5$$

↑ Whenever at least one of X, Y is zero.

So:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

$$= 0.5 - 0.8 \cdot 1$$

$$\underline{\underline{\text{Cov}(X, Y) = -0.3}}$$

$$\begin{aligned} \text{4e) } P(X=0 | X+Y=2) &= \frac{P(X=0 \cap X+Y=2)}{P(X+Y=2)} \\ &= \frac{0.3}{0.3+0.1+0.2} (= p(0,2) + p(1,1) + p(2,0)) \end{aligned}$$

$$\underline{\underline{P(X=0 | X+Y=2) = \frac{1}{2}}}$$